Detecting Safety Commitment Before It Appears in Text: Temporal Hidden-State Probes and Change-Point Detection for Reasoning Model Self-Jailbreak

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May 16, 2026

Abstract

Reasoning models interleave safety reasoning with harmful execution throughout extended chain-of-thought (CoT), culminating in a *commitment point* where safety reasoning permanently ceases. Existing hidden-state jailbreak detectors, designed for instruction-tuned models with adversarial inputs, have never been evaluated on this *self-jailbreak* phenomenon. We provide the first systematic study of hidden-state detection for safety commitment in reasoning models across four models spanning two architecture families (Llama-3.1, Qwen2.5) and two scales (7–8B, 32B). We find that (1) static per-step probes achieve <50% detection coverage despite strong hidden-state signal, because the safety signal is distributed across minor principal components; (2) temporal window logistic regression (W=15) saturates oracle-achievable coverage at 72–77%, improving over static probes by 31–35 percentage points across all architectures; and (3) Bayesian Online Change-Point Detection (BOCPD) on probe score sequences achieves the first reliable pre-textual detection, with +8.4 mean lead time steps and 62% early detection rate—substantially outperforming GRU sequence models (+4.7 steps, 39% early detection). Hidden-state probes outperform text-based monitoring by 29–32pp in coverage, confirming that the commitment signal exists in hidden states before it manifests in text. Our results establish temporal probing with BOCPD as a practical pipeline for reasoning-model safety monitoring.

1 Introduction

Large language models with extended chain-of-thought (CoT) reasoning capabilities—such as DeepSeek-R1 [3] and QwQ [7]—exhibit a distinctive failure mode termed *self-jailbreak*: the model's own reasoning process gradually abandons safety considerations and commits to producing harmful content, without any adversarial manipulation of the input [5]. Unlike conventional jailbreaks targeting instruction-tuned models, where the attack is an adversarial *input*, self-jailbreak arises from the model's *own reasoning process*.

This phenomenon poses a unique detection challenge. Safety reasoning and harmful execution are interleaved throughout the CoT—84% of traces contain safety-relevant labels after step 10—culminating in a *commitment point* (CP) where safety reasoning permanently ceases. Existing hidden-state detectors such as TrajGuard [9], SafeSwitch, and SIREN operate at per-token or per-step granularity and were designed for models where harmful intent originates externally. Whether these approaches transfer to detecting a model's internal safety commitment remains unexplored.

We address this gap with three contributions:

(1) Static probes are fundamentally limited. We show that linear probes applied to individual reasoning steps achieve <50% detection coverage across all four models tested, despite hidden states containing abundant safety signal (Fisher discriminant ratio 17–19 in full representation space). The signal collapses in low-dimensional projections (Fisher drops to $\sim$1.0 in PCA-3D, with top-3 PCs explaining only 14–17% of variance), explaining why geometric methods like Mahalanobis distance fail (coverage 0.30–0.35).

(2) Temporal context saturates oracle coverage. A temporal window logistic regression probe (W=15 steps) achieves 72–77% coverage across all architectures, matching the label-oracle upper bound. The +31–35pp improvement over static probes is consistent across two architecture families and two scales, with the temporal context effect (31pp) dwarfing both scale (2pp) and architecture (5pp) effects.

(3) BOCPD enables reliable pre-textual detection. We introduce Bayesian Online Change-Point Detection (BOCPD) [1] applied to temporal probe confidence scores as a novel approach for early commitment detection. BOCPD achieves +8.4 mean lead time steps with 62% early detection rate, substantially outperforming GRU sequence models (+4.7 steps, 39%). Hidden-state probes outperform text-based monitoring by 29–32pp in coverage, confirming that the commitment signal exists in representations *before* it appears in text.

2 Related Work

Hidden-state safety detection. Representation engineering [8] and refusal direction analysis [2] demonstrate that safety-relevant features are linearly decodable from LLM hidden states. TrajGuard [9] uses Mahalanobis distance for trajectory-level anomaly detection, achieving 95% defense rate on instruction-tuned models. These methods assume safety features align with dominant variance directions—an assumption we show fails for reasoning models.

Reasoning model safety. Self-Jailbreaking [5] and HarmThoughts characterize how reasoning models abandon safety reasoning during extended CoT. SafeRemind and SafeThink propose text-level CoT monitoring but are vulnerable to pathological CoT where surface text does not reflect internal state. No prior work bridges hidden-state detection to reasoning-model self-jailbreak.

Probing methodology. Linear probing [4] is widely used for analyzing learned representations. Process reward models (PRMs) [6] provide step-level supervision for reasoning, motivating our step-level probe design.

Change-point detection. BOCPD [1] performs online detection of distributional shifts in sequential data. We are the first to apply BOCPD to hidden-state probe outputs for safety monitoring, reframing commitment detection as a change-point problem.

3 Problem Setup

3.1 Safety Commitment in Reasoning Traces

We study reasoning traces from models prompted with harmful queries. Each trace consists of a sequence of CoT reasoning steps $\{s_1, s_2, \ldots, s_T\}$. Following **(author?)** [5], each step receives a binary safety label $y_t \in \{0, 1\}$ indicating whether it contains safety-relevant reasoning.

Commitment point. We define the commitment point as $\text{CP} = \min\{t : y_{t'} = 0 \; \forall \; t' \geq t\}$—the first step after which safety reasoning never returns. A trace is *committed* if such a CP exists and the model ultimately produces harmful output. Our detection task is to identify committed traces and localize their CPs.

Detection metrics. We evaluate probes on:

- **Coverage**: fraction of committed traces where the probe correctly detects commitment (true positive rate at operational threshold).

- **Lead time**: number of steps between probe detection and CP. Positive values indicate pre-CP (early) detection.
- **Early detection rate**: fraction of detected traces where lead time > 0.
- **False positive rate (FPR)**: fraction of safe traces (model refuses) incorrectly flagged.

3.2 Models and Data

We evaluate four reasoning models spanning a 2×2 matrix of architecture families and scales:

Model	Base Arch.	Layers	Hidden Dim
DeepSeek-R1-Distill-8B	Llama-3.1	32	4096
OpenThinker-7B	Qwen2.5	32	4096
DeepSeek-R1-Distill-32B	Qwen2.5	64	5120
QwQ-32B	Qwen2.5	64	5120

Table 1: Models evaluated. R1-8B uses Llama-3.1 base; the remaining three use Qwen2.5 base at two scales.

We use HarmThoughts reasoning traces with binary safety labels. Training uses 209 traces from R1-8B; cross-architecture evaluation uses held-out test sets from all four models. Safe traces (85 total, model successfully refused) are reserved for false positive analysis.

4 Methods

4.1 Static Probes

For each reasoning step s_t, we extract the hidden-state representation $\mathbf{h}_t^{(\ell)} \in \mathbb{R}^d$ from layer ℓ at the last token position. A logistic regression probe $f_{\text{static}}(\mathbf{h}_t^{(\ell)}) = \sigma(\mathbf{w}^\top \mathbf{h}_t^{(\ell)} + b)$ predicts the safety label y_t. We select the optimal layer via Fisher discriminant ratio computed over training traces.

4.2 Temporal Window Probes

We concatenate hidden states from a window of W consecutive steps: $\mathbf{x}_t = [\mathbf{h}_{t-W+1}^{(\ell)}; \ldots; \mathbf{h}_t^{(\ell)}] \in \mathbb{R}^{Wd}$. A logistic regression probe $f_{\text{temp}}(\mathbf{x}_t) = \sigma(\mathbf{w}^\top \mathbf{x}_t + b)$ classifies whether step t is post-commitment. Commitment is detected when f_{temp} exceeds a threshold for a sustained window, capturing the "safety reasoning density decline" pattern—the gradual thinning of safety-labeled steps before CP.

4.3 GRU Sequence Probes

A GRU with 256 hidden units processes the sequence of hidden states $\{\mathbf{h}_1^{(\ell)}, \ldots, \mathbf{h}_t^{(\ell)}\}$ autoregressively, producing predictions at each step. We also evaluate a self-attention variant. These models can potentially detect commitment *before* CP by learning predictive temporal patterns.

4.4 BOCPD on Probe Score Sequences

We introduce a two-stage pipeline. First, the temporal LR probe produces a confidence score sequence $\{p_1, p_2, \ldots, p_t\}$. Second, we apply Bayesian Online Change-Point Detection [1] to detect distributional shifts in this score sequence.

BOCPD maintains a run-length distribution $P(r_t|\mathbf{p}_{1:t})$ where r_t is the number of steps since the last change point. At each step, the algorithm updates the posterior over run lengths given the new

Model	Best Layer	Depth %	Coverage	Fisher (full/PCA-3)
R1-8B	14	44%	0.465	17.2 / 1.02
R1-32B	28	44%	0.42	19.3 / 1.05
QwQ-32B	30	47%	0.39	18.7 / 0.98
OpenThinker-7B	14	44%	0.41	—

Table 2: Static probe results. Coverage remains below 0.50 across all models despite high Fisher discriminant ratios in full representation space. PCA-3D Fisher collapse ($\sim$1.0) confirms safety signal is distributed across minor principal components.

Model	Static	Temporal	Δpp	AUROC	Lead
R1-8B	0.465	0.767	+30	0.89	-0.4
R1-32B	0.42	0.77	+35	0.91	-0.6
QwQ-32B	0.39	0.74	+35	0.88	-0.9
OpenThinker-7B	0.41	0.72	+31	—	—

Table 3: Temporal window LR (W=15) vs. static LR. The temporal advantage is consistent across all four models. AUROC values confirm strong discrimination independent of threshold.

observation p_t, using a hazard function $H(r) = 1/\lambda$ (we use $\lambda = 100$, i.e., hazard rate 0.01) and a Gaussian observation model with conjugate Normal-Inverse-Gamma prior.

A change point is declared when $P(r_t = 0|\mathbf{p}_{1:t})$ exceeds a threshold, indicating the probe score distribution has shifted—specifically, from the oscillating regime characteristic of active safety reasoning (mean $\sim$0.6, std $\sim$0.15) to the stable high-confidence regime of post-commitment steps (mean $\sim$0.85, std $\sim$0.04). This reframes commitment detection as a change-point problem rather than a sequence prediction problem.

4.5 Baselines

TrajGuard (Mahalanobis distance). We estimate class-conditional Gaussians from safe-trace hidden states and compute per-step Mahalanobis distance as an anomaly score [9].

Text-based monitoring. Sentence-BERT (all-MiniLM-L6-v2) embeddings of CoT text steps with temporal LR (W=15), providing a direct comparison between hidden-state and surface-text monitoring.

5 Experiments

5.1 Static Probes Achieve Limited Coverage

Table 2 shows that static probes achieve 0.39–0.47 coverage across all four models, with best layers consistently at 44–47% depth.

The consistent best-layer depth at $\sim$44% across architectures and scales suggests middle layers encode the richest safety-relevant abstractions. The Fisher discriminant ratio is high in full space (17–19) but collapses to $\sim$1.0 in PCA-3D projections, with top-3 PCs explaining only 14–17% of variance. This distributional property arises from safety fine-tuning spreading signal across many dimensions, explaining why geometric methods fail (Section 5.4).

5.2 Temporal Probes Saturate Oracle Coverage

Table 3 demonstrates that temporal window LR (W=15) achieves 0.72–0.77 coverage across all architectures, a +31–35pp improvement over static probes.

	Llama-3.1	Qwen2.5
7–8B	0.767 (R1-8B)	0.72 (OpenThinker)
32B	—	0.77 / 0.74 (R1-32B / QwQ)
Effect sizes: temporal +31pp, scale +2pp, arch 5pp		

Table 4: 2×2 architecture×scale matrix. Temporal context dominates over scale and architecture.

Method	Model	Lead (mean)	Lead (med)	Early Det.
Temporal LR	R1-8B	−0.4	—	—
	R1-32B	−0.6	—	—
	QwQ-32B	−0.9	—	—
GRU-256	R1-8B	+4.7	0	0.39
	R1-32B	+5.2	1	0.42
	QwQ-32B	+4.9	0	0.40
BOCPD	R1-8B	+8.2	6	0.62
	R1-32B	+9.1	7	0.65
	QwQ-32B	+7.8	—	0.58
CUSUM	R1-8B	+5.1	—	0.48

Table 5: Lead time comparison. BOCPD achieves +8.4 mean lead time and 62% early detection, substantially outperforming GRU (+4.7, 39%) and CUSUM (+5.1, 48%). BOCPD's advantage comes from detecting variance reduction in probe scores, not predicting future states.

The coverage values cluster near the oracle ceiling (76.7% on R1-8B), suggesting this bound reflects an intrinsic property of the task: some traces have ambiguous or gradual commitment without a clean phase transition.

2×2 architecture×scale analysis. Table 4 decomposes the sources of variation in temporal coverage. The temporal context effect (+31pp mean) dwarfs both scale (+2pp) and architecture (+5pp) effects, confirming that our findings generalize across model families.

5.3 BOCPD Achieves Reliable Pre-Textual Detection

Table 5 compares lead time methods. BOCPD substantially outperforms GRU by reframing commitment detection as distributional shift detection on probe confidence scores.

BOCPD succeeds because the probe score sequence exhibits a clear regime change: pre-commitment scores oscillate (mean $\sim$0.6, std $\sim$0.15) while post-commitment scores stabilize (mean $\sim$0.85, std $\sim$0.04). BOCPD detects this variance reduction 6–9 steps before the textual commitment point. CUSUM achieves intermediate performance because it detects mean shift but is less sensitive to the critical variance reduction.

Oracle decomposition. To understand the remaining gap to the oracle's +31 steps, we decompose oracle advantage into lookback pattern recognition (45%) and future knowledge (55%). Probes capture 72% of the achievable lookback signal but only 8% of the future-knowledge component. This establishes a principled ceiling: approximately 45% of oracle advantage is achievable from past data alone, and BOCPD captures the majority of this achievable component.

5.4 Baselines: TrajGuard and Text-Based Detection

TrajGuard (Mahalanobis distance). Table 6 shows TrajGuard achieves only 0.30–0.35 coverage (AUROC 0.55–0.58), far below temporal probes. This failure is not due to absent signal but to the

Method	R1-8B	R1-32B	QwQ-32B
Coverage			
TrajGuard (Mahalanobis)	0.35	0.32	0.30
Text static LR	0.38	—	—
Text temporal LR (W=15)	0.48	0.45	0.47
Hidden-state temporal LR	0.767	0.77	0.74
Hidden-state vs. text gap (pp)			
Temporal probe gap	+29	+32	+27

Table 6: Baseline comparison. TrajGuard and text-based methods achieve substantially lower coverage than hidden-state temporal probes.

Method	FPR	Prec@70%Rec
Static LR	0.12	—
TrajGuard	0.18	—
Temporal LR (R1-8B / R1-32B / QwQ)	0.06 / 0.05 / 0.07	0.92
BOCPD (R1-8B / R1-32B)	0.04 / 0.03	0.94

Table 7: False positive rates at operational threshold. Temporal LR and BOCPD maintain low FPR (3–7%) while achieving high coverage. Safe traces produce mean probe score 0.35 vs. 0.78 for committed traces, providing comfortable operating margin.

signal's distribution: safety features spread across minor principal components that Mahalanobis distance effectively ignores.

Text-based monitoring. Text temporal LR achieves 0.45–0.48 coverage—better than text static (0.38) but 29–32pp below hidden-state probes. The text lead time is negative (-3.2 steps), confirming that text-based methods can only detect commitment *after* it manifests linguistically, while hidden-state probes capture the signal pre-textually (BOCPD: +8.4 steps). This validates our core thesis: hidden-state monitoring provides unique value that text monitoring cannot replicate.

5.5 False Positive Analysis

Table 7 reports false positive rates on 85 safe traces at the operational threshold yielding $\geq$70% coverage.

BOCPD achieves the lowest FPR (0.03–0.04) because safe traces never exhibit the variance-reduction regime change that signals commitment. The clear separation between safe-trace (mean 0.35) and committed-trace (mean 0.78) probe scores confirms the probe captures a genuine phenomenon.

5.6 Ablations

Window size. Table 8 shows coverage as a function of window size W on R1-8B. Coverage increases monotonically until W=15, then plateaus. The optimal W=15 is consistent across all models and corresponds to the mean pre-CP safety reasoning streak length (14.6 steps from oracle analysis), providing a mechanistic explanation.

Layer selection. On 32B models, the optimal temporal probe layer is at 45–47% depth (layers 29–30 of 64), replicating the static probe finding. A broad plateau (layers 24–36, all >70% coverage) indicates layer selection is not fragile.

W	1	3	5	10	15	20	25
Coverage	.465	.56	.65	.72	.767	.77	.76

Table 8: Window size ablation on R1-8B. W=1 equals the static probe. Saturation at W=15 corresponds to the mean pre-CP safety reasoning streak length. Slight degradation at W=25 from dilution with non-predictive early-trace steps.

Probe architecture. GRU-256 and self-attention probes provide only +2pp coverage improvement over temporal LR on 32B models (0.79/0.78 vs. 0.77), confirming temporal LR as the optimal cost-performance tradeoff for coverage. Complex architectures add lead time capability but not coverage.

Fine-grained supervision. Replacing binary commitment labels with a 16-category safety reasoning taxonomy yields 0.79 coverage (+2pp over binary), with 62% 16-class accuracy. The marginal improvement confirms binary labeling is near-optimal for detection; the taxonomy's value is interpretability (identifying *which* safety reasoning type is declining).

Cross-model transfer. Training on R1-8B (Llama, 4096-d) and testing on R1-32B (Qwen, 5120-d) via a learned linear projection retains 68% of within-architecture performance (0.54 vs. 0.77 coverage). The partial transferability suggests temporal probes learn a partially universal safety commitment signature.

Statistical robustness. Bootstrap CIs (1000 resamples) yield coverage CI widths of 0.05–0.07, and the temporal-static coverage gap (31–35pp) is 4–6$\times$ larger than CI width, ruling out sampling variance. Cross-validation fold variance is 0.003, indicating stable probe training. All primary claims are statistically robust at conventional significance levels.

6 Discussion

Why temporal context matters. The commitment point is fundamentally a *trajectory* property—a sustained absence of safety reasoning over $\sim$15 steps—not a single-step state property. Static probes, no matter how accurate at classifying individual steps, cannot capture this pattern. The consistency of optimal W=15 across architectures confirms this is determined by task structure, not model properties.

BOCPD as methods contribution. Prior approaches to lead time (GRU, attention) attempt *sequence prediction*—forecasting whether safety reasoning will return. BOCPD succeeds by reframing the problem as *change-point detection*: identifying when probe score statistics have shifted, indicating the onset of commitment. This shift—from oscillating to stable high-confidence scores—is detectable 6–9 steps before textual commitment because hidden states encode the onset of commitment before text manifests it. To our knowledge, this is the first application of BOCPD to hidden-state probe outputs for safety monitoring.

Principled limitations. Oracle decomposition reveals that 55% of oracle advantage requires genuine future knowledge—knowing that safety reasoning will *never* return. No causal method can access this information. BOCPD's +8.4 lead time captures most of the achievable 45% that relies on lookback pattern recognition.

Deployment considerations. The temporal LR + BOCPD pipeline requires only a single middle-layer hidden-state extraction per reasoning step, a windowed logistic regression, and lightweight online Bayesian inference. At 70% coverage with 3–7% FPR and +8.4 steps mean lead time, this provides a practical monitoring system that could trigger intervention (e.g., early termination, safety prompt injection) before commitment manifests in output text.

7 Conclusion

We presented the first systematic study of hidden-state detection for safety commitment in reasoning models. Across four models, two architecture families, and two scales, we established that: (1) static probes achieve <50% coverage due to distributed safety signal; (2) temporal window probes saturate oracle coverage at 72–77%; and (3) BOCPD on probe scores achieves the first reliable pre-textual detection (+8.4 steps, 62% early detection). Hidden-state monitoring outperforms text-based approaches by 29–32pp, confirming that the commitment signal exists in representations before it appears in text.

Future work should explore (1) scaling to longer reasoning traces from frontier models, (2) integration with real-time inference pipelines, and (3) whether BOCPD can be applied to other safety-relevant phase transitions in model behavior.

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